Assignment Brief

BEMM462 Operations Analytics Term 3 –

2024/25 Instructor: Dr. Shazib Shaikh

Formative Assessments:

In-class Exercises

The Formative Assessments are weekly exercises conducted during your Seminars. Exercises will become available before the sessions in the related weekly ELE page. Feedback will be provided verbally during these sessions and do not count towards your module grade. You are encouraged to attempt all exercises and participate in discussions to enhance your understanding.

Summative Assessments:

The Summative Assessments are assessed by group coursework exercise worth 30% and individual report worth 70% of your final mark. Below is your individual report.

Individual Report Assignment

Length: 2000 words

Weight: 70% of the overall module mark

Deadline: 30 June, 2025 (Monday) at 12:00 noon via ELE2

Submission Requirements: Submissions must be through ELE2. I won't be able to accept submissions personally through emails. Please give yourself enough time to upload the documents and meet the deadline.

Important notice: This summative assessment is AI-Prohibited. GenAI tools must not be used in the completion of this assessment.

Ensure you include the School provided cover page declaration regarding GenAI usage

Summative Assessment 2: Individual Report (70%)

You have recently been appointed as a management consultant to provide solutions to the following three projects. Neatly type your answers on a Word document and convert it to PDF before sub- mitting. The sub-items for the problems are important to be addressed and serve as a guide to the more open-ended questions. Please ensure that your report includes all diagrams drawn, equations, outputs of analysis or screenshots from any software used (e.g. Excel/R/Python etc). The report should provide ***detailed*** *explanations, examples, and workings on how you derive the answers to justify the marks allocated for each question. Any cited references in the APA format should be included in your report. Please include all software codes in your submission.*

Project 1

A small engineering consulting firm is establishing a plan for next year. The director and the three partners need to prioritize the eight projects. The expected profit for each project is given in the following table together with the number of person-days required to prepare each project. Another measure included is the computing resource measured as the computer processing unit (CPU) time (in hours). You have been tasked to evaluate which projects they ought to pursue based on efficiency. Given the sample data below, you plan to construct a data envelopment analysis (DEA) model:

Project	Profit (in million £)	person-days	CPU time (hours)
1	2.1	550	200
2	0.5	400	150
3	3	300	400
4	2	350	450
5	1	450	300
6	1.5	500	150
7	0.6	350	200
8	1.8	200	600

1. Which measures would you regard as inputs and outputs, respectively? Please provide your justification. .... (5 marks)

2. Conduct data envelopment analysis, explain what type of DEA model is selected and describe the project you regard as "efficient" and "inefficient", respectively. .... (8 marks)

3. Which project would you recommend and why? .... (7 marks)

4. In what ways can you improve your recommendation? You are free to openly source for other datasets (or create one) and justify the new set of input and output measures before conducting another data envelopment analysis.

   Elaborate on your recommendations and solutions. .... (10 marks)

5. Assuming you have to consider a project from a different industry in your analysis and you can have any data and measurement at your disposal, what would be the ideal steps taken to ensure that the projects from different industries are comparable?........................ (10 marks)

(Total: 40 marks)

Project 2

A company has a total of 3 factories to produce steel and iron. Every month, the company must meet the following demands: 3200 tonnes of steel and 1000 tonnes of iron. Each factory has a different shipping cost per tonne (£/tonne) for each product as shown in the table below:

Factory	Steel	Iron
1	200	500
2	800	400
3	500	1000

The warehouses in Factory 1, Factory 2 and Factory 3 can store up to 2000 tonnes,1500 tonnes and 2500 tonnes of products, respectively.

1. Based on the table above, determine the optimal monthly shipment plan. That is, how many tonnes does each factory need to ship to fulfil the demand? The company seeks a solution that would minimise the shipping cost. .... (5 marks)

2. Suppose Factory 3 will be unavailable for the next few months. Assuming the same demand levels, will the company able to fulfil that demand? How would demand best be allocated between Factory 1 and Factory 2 when Factory 3 is non-operational?.................... (6 marks)

3. The cost of raw materials in each factory is shown in the table below. Change the company objective function to minimize all costs and find the new optimal assignment plan. Does the new solution change from the previous one and explain whether that outcome is predictable . ... (9 marks)

Factory	Steel raw material cost	Iron raw material cost
1	50	100
2	70	120
3	45	130

(Total: 20 marks)

Project 3

Please attempt to resolve Tasks 1 and 2 aided by optimization software (e.g. Excel Solver). Ensure to demonstrate your analysis by sharing your results and outputs and explaining relevant sections

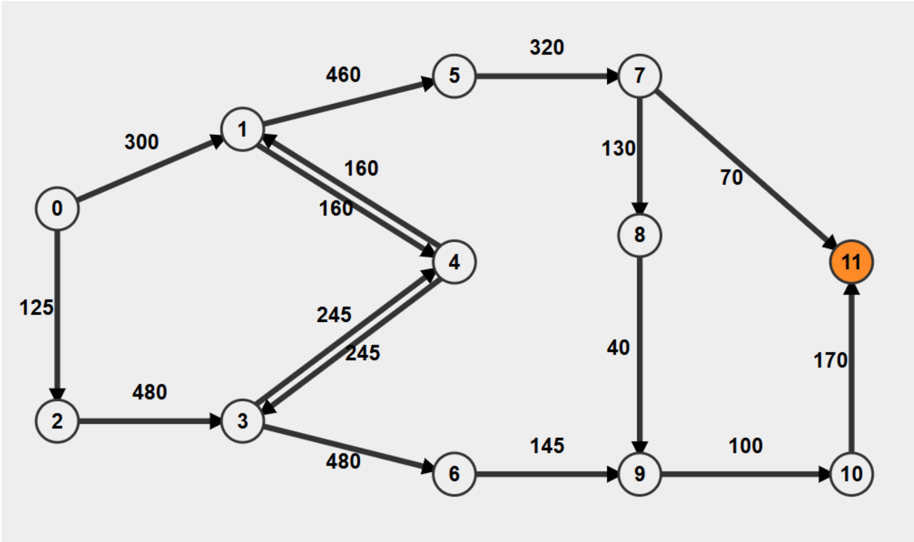

Figure 1 - A network model - see how each task interprets this diagram differently

1. Consider Figure 1 to be a social media influencer network: the nodes are the influencers, the arrows are direction of influence, the values on the lines are the cost of investing in that influence. For the brand you are representing to remain viral, your digital marketing agency needs to maintain a flow of connection from node 0 to node 11. Which route of influence flow would you recommend to your client and why? .... (10 marks)

2. Now consider Figure 1 to be a network model for a multinational game development company: the nodes are globally distributed development teams, the arrows are data transfer network links (mostly in one direction to disperse from one time zone to the next), the values on the lines represent bandwidth (larger values represent larger "pipes" through which data can flow). The rollout of their large game builds must begin at Node 0 (earliest time zone and development base) and then be synched and transferred with the maximum amount of build data getting from Node 0 to Node 11 (latest time zone). Nodes missed in this rollout to Node 11 are less priority and can synch later. But it is crucial that the maximum rollout happens at first upload from Node 0 to Node 11.

    a. What would be the optimal rollout sequence to ensure the maximum amount of data on first rollout from Node 0 reach Node 11? .... (9 marks)

    b. Can we identify any major bottlenecks to this first rollout? .... (3 marks)

3. From your newfound knowledge of network optimisation, list and describe how shortest path and maximum flow models have actual applications in real-world business problems. Please provide at least 2 applications for each type of model. You should NOT use the application listed in (1) and (2) above. ....(18 marks)

(Total: 40 marks)

Marking Grid

Mark	<50	50 - 59	60 – 69	≥70	Weighting
ASSESSMENT CATEGORIES	Fail	Pass	Merit	Distinction	% of total mark
KNOWLEDGE & UNDER-STANDING OF SUBJECT	Demonstrates little knowledge or understanding of the field; demon-strates significant weaknesses in the knowledge base, and/or sim- ply reproduces knowledge without evidence of under-standing	Demonstrates a sound knowledge and understanding of material within a specialised field of study; demonstrates an understanding of current theoretical and methodolog- ical approaches and how these affect the way the knowledge base is interpreted	Produces work with a well-defined focus; demon-strates a systematic knowledge, un-derstanding and critical awareness of current prob-lems and/or new insights, much of which is at, or informed by, the forefront of the academic discipline, field of study or area of professional practice	Produces work of exceptional standard, reflect-ing outstanding knowledge and understanding of material, Displays exceptional mas-tery of a complex and specialised area of knowledge and skills, with an exceptional critical awareness of current prob-lems and/or new insights at the forefront of the field	40%
ANALYSIS	Analytic approach is wrong, incor-rectly applied, or incorrectly explained	Analytic approach is inappropriate, understanding of the method is not demonstrated, and implications are not correctly understood or interpreted	Appropriate an-alytic approach, but some elements of the method are not explained, and important implica-tions are missed	Analytic approach is appropriate, cor- rectly explained in detail, and the im-plications are well understood	30%
VISUALISATION	Visualisation is completely unsuited to the data, very poorly executed, and it is impossible to understand	Visualisation is not appropriate for the data, execution is poor, and it's un-clear what its pur-pose is	Visualisation is ap-propriate, but it is not well executed and requires addi-tional explanation	Visualisation use is appropriate, clean and clearly explained, it stands on its own, and has a quick clear meaning	20%
STYLE, STRUC-TURE & KEY COMPONENTS	Poor structure and grammar which is hard to follow or understand; incorrectly for-matted; arguments lack clarity. Poor referencing	A reasonable structure and use of language. Some errors in grammar, vocabulary and referencing. Clear arguments and display of logic in construction. Referencing con-ventions generally followed	Good structure and grammar, which is easy to follow and understand. Few instances of typos or formatting errors. The as-signment is clear, logically struc-tured, well written and referencing conventions are respected.	Clearly structured and lucidly ex-pressed. Minor or no errors in lan-guage, grammar or referencing. All aspects of the assignment are delivered o an exceptional degree of clarity, structure and detail.	10%
					Total 100%